

Instructor: Date:

Learner: Regional Branch:

Program: Center:

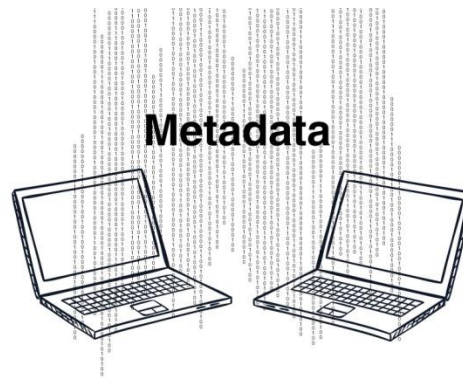
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Let's Start!

I think I need to locate and change some information of this database...



Learning Outcome:

Exchange written and spoken information on principles, concepts, data base objects related to relational database management system in spoken or written texts.
Describe relational database, its concepts, and principles in spoken or written texts.

Can Do Statements:

- I can understand written information about independence in database systems.
- I can identify information about logical and physical data independence.
- I can write a summary about a text on data independence

Ask your partner:

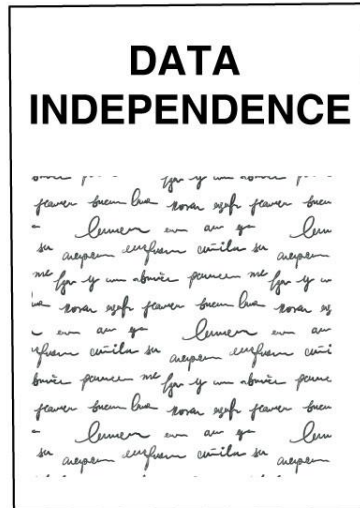
- What do you have to consider before you edit the information of a database?
Share your thoughts.
- What do you have to consider after you edit the information of a database?
Share your thoughts.





Let's Explore!

An article on data independence



Instructions: Read the text below.

Data independence is a very important aspect related to DBMS or Database Management System, as it is key to manage and edit the information we can find in a database. Now, it is important to clarify that databases should follow a multi-layered architecture, which enables programmers to easily make changes in specific layers of the database systems without affecting the information contained in the rest of it. In fact, with data independence making and keeping these databases updated is not as hard as many people think as they can actually use metadata which, in simple words, is the data we can obtain out of preexisting data to retrieve and locate information efficiently. In fact, satisfying the users' requirements is not as tough as it may seem.

A colleague of mine didn't know anything about data independence that is why I told him that he had to check two different mechanisms: logical data and physical data independence. I told him that when we referred to the logical data independence mechanism, we didn't refer to the way data was stored in the inside of the database, but to the way it was managed. The logical data independence does not change the data that is contained on the disk, but it changes the information we store on the database's table formats.

On the other hand, we have the physical data independence, which is not as confusing as the previous mechanism. With this one, we can change the physical data of the database without affecting its logical data or schema of the system. This is useful when there is not as much storage capacity as required, so we decide to upgrade it without affecting the overall core system's schema.





Let's Practice!

Independence in Database Systems

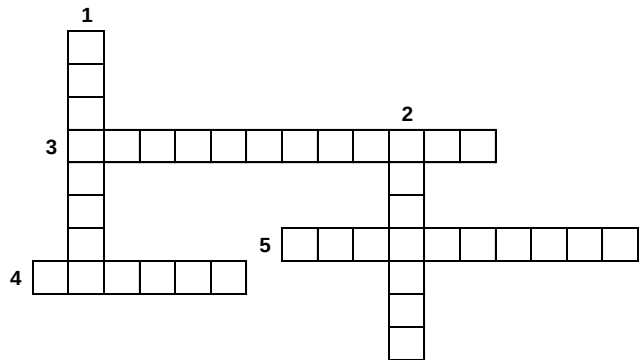
Instructions: Read the clues. Complete the crossword.

DOWN

1. The information we get out of the other data contained in the database.
2. When the information of a database is current and valid, we can say that the database is __ d ____.

ACROSS

3. It is advisable to create a database that contains a multi-layered ar ____.
4. With data independence, when you modify the information in one of the ____ e ____ of the system, that change does not affect the whole system.
5. DBMS stands for database ____ ____ ____ system.



Let's Practice!

Once you read about data independence, the topic is not as tough as you thought.

Instructions: Read the question. Select the correct answer.

1. Understanding data independence is not _____ it seems.
 - a. as confuse as
 - b. as confused as
 - c. as confusing as
 - d. as confusing us
2. When databases have a multilayered architecture, modifying their data is _____ many believe.
 - a. not as easy as
 - b. not as hard as
 - c. not as hard like
 - d. as amazing as



**Let's Practice!**

3. Dependent data is not _____ some programmers think as it is more difficult to make any changes in that type of database.
 - a. us needed us
 - b. as need as
 - c. as needing as
 - d. as needed as
4. Metadata is _____ she thinks; it helps to retrieve information faster and easier.
 - a. as broken as
 - b. not as important as
 - c. not as useless as
 - d. not as useful as
5. Modifying information in OOP _____ in some other programming languages thanks to the data independence.
 - a. is not as difficult as
 - b. is not us difficult as
 - c. is not as difficult us
 - d. is not as difficult

**Extension Activity**

Let's learn more about data independence.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- a) Look for more information about data independence and metadata.
- b) Find an article that develops those topics.
- c) Make a short summary of the article you read including only the information you consider important.
- d) Share your summary with a partner.

Self-Assessment**Yes****No**

1. I look for more information about data independence and metadata.
2. I find an article that develops the topics of data independence and metadata.
3. I make a short summary of the important information of the text I found.
4. I share my summary with a classmate.

☐	☐
☐	☐
☐	☐
☐	☐



Target Words

- **metadata** (noun): information obtained from data gathering
e.g. In fact, with data independence making and keeping these databases updated is not as hard as many people think as they can actually use metadata which, in simple words, is the data we can obtain out of pre-existing data to retrieve and locate information efficiently.
- **locate** (verb): to find the current position of something or someone
e.g. They can actually use metadata which, in simple words, is the data we can obtain out of pre-existing data to retrieve and locate information efficiently.
- **satisfy** (verb): to fulfill someone's expectations
e.g. In fact, satisfying the users' requirements is not as tough as it may seem.
- **tough** (adjective): difficult, hard to do
e.g. In fact, satisfying the users' requirements is not as tough as it may seem.
- **updated** (adjective): with current and valid information
e.g. In fact, with data independence making and keeping these databases updated is not as hard as many people think.



Learning Points

Comparisons with Not as... As

We can make comparisons with the structure not as... as. We use it to make comparisons between two different elements using adjectives.

When we use the expression not as... as, we say that the two elements we are referring to are not equal. Plus, we say that the first element in the comparison is less than the second one.

Check the following structure:

Element 1 + verb 'to be' + not as + adjective (+ complement) + as + element 2 (+ complement).

For example:

The logical schema **is not as easy to understand as** the physical schema.
Mary **is not as confused** with the independence in DBMS **as** Gabriel.

Examples

- *In fact, satisfying the users' requirements is not as tough as it may seem.*
- *This is useful when there is not as much storage capacity as required, so we decide to upgrade it without affecting the overall core system's schema.*





Learning Points

Review of Simple Tenses: Negative Form

We use the **present simple** and **past simple** to express different ideas or events in different times.

Present Simple

Remember that we use the present simple tense to talk about habits or facts. In this case, we will review its negative form. In order to write negative statements using this tense, we follow the structure:

Subject + **do/does** + **not** + verb in base form + complement.

For example:

*The company **does not satisfy** the requirements of the user.*

*They **do not understand** what data independence is.*

Remember that we can also use the verb 'to be' in the present simple tense.

For example:

*She **is not** ready to work for this company.*

*They **are not** at the meeting.*

Past Simple

We use the past simple to refer to events or actions that started and finished in the past. In other words, they are no longer happening. In order to write negative sentences using the past simple, we follow the structure:

Subject + **did** + **not** + **verb in base form** + complement.

For example:

*The company **did not satisfy** the requirements of the user.*

*They **did not understand** what data independence is.*

Remember that we can also use the verb 'to be' in the past simple tense.

For example:

*She **was not** ready to work for this company.*

*They **were not** at the meeting.*

Examples

- *The logical data independence does not change the data that is contained on the disk, but it changes the information we store on the database's table formats.*
- *I told him that when we referred to the logical data independence mechanism, we didn't refer to the way data was stored in the inside of the database, but to the way it was managed.*

